

FLOTATION BENCH // v2 – anisotropy + depressant model

rule-based surface chemistry reasoning – not a DFT solver

MINERAL SURFACE

Gibbsite (Al(OH)₃)

COLLECTOR

Fatty acid (e.g. oleic acid / sunflower-oil derived)

DEPRESSANT (OPTIONAL)

None

DEPRESSANT DOSE (RELATIVE)

0

PH (PULP CONDITIONS)

7.0

CLASS	Bauxite – aluminum hydroxide
PZC	pH 9
SURFACE SITE	Al-OH
HSAB	hard
Dominant phase in gibbsitic bauxite. Surface Al-OH amphoteric groups.	

Models known HSAB / electrostatic / competitive-adsorption rules. Not a substitute for zeta-potential measurement, IR spectroscopy of adsorbed species, or bench-scale flotation on your actual sample.

SURFACE – COLLECTOR BONDING, LIVE

confidence: medium

drag to rotate

chemisorption (chelation)

STRONG

Fatty acid (e.g. oleic acid / sunflower-oil derived)'s -COO^- head group chelates onto Al-OH sites on Gibbsite. Surface is positive at pH 7 (PZC 9).

SURFACE CHARGE

positive

Δ FROM PZC

2.0 units

HSAB PAIRING

hard / hard

Oleic acid is documented as workable for gibbsite flotation (Bittencourt 1989), but separate bench data using a different collector class (CTAB) measured only 37% gibbsite recovery under nominally favorable conditions — suggests recovery ceilings this rule engine doesn't quantify.

Basis: Bittencourt (1989),